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BRANCH : CSM

SECTION : B

COURSE NAME : FULL STACK WEB DEVELOPMENT

CODE : 23CM4121

TITLE OF THE EXPERIMENT : To create a JavaScript-based student profile application using classes, objects, DOM manipulation, dynamic HTML elements, and event handling.

DESCRIPTION:

The Student Profile application is a simple web application developed using HTML, CSS, and JavaScript. It uses a JavaScript Student class with properties such as Name, Roll Number, Department, and CGPA. The user enters the student details through input fields, and when the Display Student Profile button is clicked, JavaScript collects the entered values using DOM methods. A Student object is then created using the class constructor. The program dynamically creates HTML elements using `createElement()`, adds student information using `textContent`, and displays the elements on the webpage using `appendChild()`. The `addEventListener()` method is used to handle the button-click event. Basic CSS is applied to make the dynamically generated student profile clear and attractive. Thus, the program demonstrates JavaScript classes and objects, DOM selection, dynamic HTML element creation, content modification, DOM manipulation, and event handling.

PROGRAM :

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Profile</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      text-align: center;
      padding: 30px;
    }

    .container {
      background-color: white;
      width: 400px;
      margin: auto;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 0 10px gray;
    }
  </style>
</head>
<body>
```

```

    input {
        width: 90%;
        padding: 10px;
        margin: 8px;
        box-sizing: border-box;
        border: 1px solid #ccc;
        border-radius: 5px;
    }

    /* Updated Button Styling */
    button {
        padding: 12px 24px;
        background-color: #0f766e; /* Deep Teal */
        color: #ffffff;
        font-size: 15px;
        font-weight: 600;
        border: none;
        border-radius: 6px;
        cursor: pointer;
        box-shadow: 0 4px 6px -1px rgba(15, 118, 110, 0.3);
        transition: background-color 0.2s ease, transform 0.1s ease, box-shadow 0.2s
ease;
    }

    button:hover {
        background-color: #115e59; /* Darker Teal on hover */
        box-shadow: 0 6px 8px -1px rgba(15, 118, 110, 0.4);
    }

    button:active {
        transform: scale(0.98);
    }

    #profile {
        margin-top: 20px;
        padding: 20px;
        background-color: #e8f0fe;
        border-radius: 10px;
        text-align: left;
    }

    #profile h2 {
        text-align: center;
        margin-top: 0;
    }
</style>
</head>

<body>

    <div class="container">

        <h1>Student Profile</h1>

```

```

<input type="text" id="name" placeholder="Enter Name">
<input type="text" id="roll" placeholder="Enter Roll Number">
<input type="text" id="department" placeholder="Enter Department">
<input type="number" id="cgpa" step="0.01" placeholder="Enter CGPA">

<br><br>

<button id="displayBtn">Display Student Profile</button>

<div id="profile"></div>

</div>

<script>
    // Creating Student class
    class Student {
        constructor(name, rollNumber, department, cgpa) {
            this.name = name;
            this.rollNumber = rollNumber;
            this.department = department;
            this.cgpa = cgpa;
        }
    }

    // Select button using DOM
    const button = document.getElementById("displayBtn");

    // Event handling
    button.addEventListener("click", function () {
        // Get values from input fields
        let name = document.getElementById("name").value;
        let roll = document.getElementById("roll").value;
        let department = document.getElementById("department").value;
        let cgpa = document.getElementById("cgpa").value;

        // Create Student object
        let student = new Student(name, roll, department, cgpa);

        // Select profile area
        let profile = document.getElementById("profile");

        // Clear previous content
        profile.innerHTML = "";

        // Create elements dynamically
        let heading = document.createElement("h2");
        heading.textContent = "Student Profile";

        let nameElement = document.createElement("p");
        nameElement.textContent = "Name          : " + student.name;

        let rollElement = document.createElement("p");

```

```

        rollElement.textContent = "Roll No      : " + student.rollNumber;

        let departmentElement = document.createElement("p");
        departmentElement.textContent = "Department : " + student.department;

        let cgpaElement = document.createElement("p");
        cgpaElement.textContent = "CGPA          : " + student.cgpa;

        // Add elements to webpage
        profile.appendChild(heading);
        profile.appendChild(nameElement);
        profile.appendChild(rollElement);
        profile.appendChild(departmentElement);
        profile.appendChild(cgpaElement);
    });
</script>

</body>
</html>

```

OUTPUT :

The screenshot shows a web browser window titled 'Student Profile'. The address bar indicates the file path 'C:/Users/black/Desktop/fso_2.html'. The main content area displays a form with four input fields containing the following values: 'Rowthu Bhaskar Abhiram', 'A24126552114', 'CSM', and '8.6'. Below these fields is a green button labeled 'Display Student Profile'. Below the button, the output of the form is displayed in a light blue box with the title 'Student Profile'. The output shows the following details: 'Name : Rowthu Bhaskar Abhiram', 'Roll No : A24126552114', 'Department : CSM', and 'CGPA : 8.6'.